

Ajith Kanumuri

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EDUCATION

Master of Computer Science | North Carolina State University | Raleigh, NC, USA | May 2026

Post Graduate Diploma in Artificial Intelligence | CDAC ACTS | Pune, India | Feb 2021

Bachelor of Technology in Computer Science and Engineering | Vellore Institute of Technology | Vellore, India | Jul 2019

SKILLS

Languages: Python, SQL | **ML/DL:** PyTorch, TensorFlow, scikit-learn, Keras, OpenCV

Data/Visualization: Pandas, NumPy, Matplotlib, Plotly | **GenAI:** LLMs, Prompt Engineering, RAG, Vector Search, MCP, Agents

Cloud: GCP (Vertex AI, BigQuery, Cloud Storage, Dialogflow CX), AWS (SageMaker, Step Functions, ECR, S3)

Databases: PostgreSQL, BigQuery, SingleStore (MemSQL), MySQL/MariaDB

Infra/Dev: Docker, Kubernetes, Git (GitHub/Bitbucket) | **MLOps:** MLFlow, CometML, DVC, Kubeflow, ZenML

EXPERIENCE

Graduate Teaching Assistant | NCSU | Raleigh, NC | Aug 2025 – May 2026

- Supported students and professors for a Python course in Fall 2025 and for a DBMS course in Spring 2026.
- Held weekly office hours and resolved questions and unblocked students promptly.
- Graded 560 projects and assignments with standardized rubrics and returned detailed feedback within 10 days.
- Maintained course materials and the gradebook in Moodle and ensured accurate, up-to-date records.
- Designed 3 graded projects and assignments with clear specs and evaluation criteria.

Machine Learning Researcher | NCSU | Raleigh, NC | Sep 2024 – Nov 2024

- Parallelized inference modules in a research ML pipeline, improving throughput by 10%.

Software Engineer/Data Scientist | ElastiQ | Remote, India | Dec 2023 – Jul 2024

- Built PCB component-presence detection (CV) on GCP; achieved ~99% accuracy via balanced dataset curation and iterative model improvements.
- Reduced a data pipeline's execution time by ~95% by parallelizing architecture and re-implementing on AWS SageMaker, Step Functions, ECR, and Python.
- Improved a time-series forecasting model by ~14% RMSE through feature engineering and model selection.

Data Scientist/Machine Learning Engineer | Mavenir | Bengaluru, India | Mar 2021 – Oct 2023

Promoted from Graduate Engineer to Member of Technical Staff I (R&D)

- Reduced invalid data points by ~90% in a data pipeline using Shell, Python, SQL, AWS, improving data integrity.
- Removed preprocessing bottlenecks by vectorizing logic and reduced runtime from 12 hours to 40 seconds (~×1000).
- Presented a PoC to 10+ clients; featured as 1 of 7 key demos at O-RAN Plugfest 2021.
- Led Deep RL experiments across 5+ use cases (TS, ES, etc), improving output KPIs by 28% and cutting network overhead by 11.5%.
- Coordinated across 8 DS/ML engineers, 4 data engineers, and 3 domain experts to deliver milestones.
- Contributed to work leading to an IEEE TMC publication (150+ citations).

PROJECTS

GenAI for SE Course Project | Sep 2025 – Dec 2025 | GitHub: <https://github.com/GenAI4GenZ/SmartNote-2.0>

- Implemented an automated release note generator based on commit messages, code changes and pull requests with 85.3% commit coverage and 20% lesser token usage compared to state-of-the-art.

Efficient Deep Learning Course Project | Jan 2025 – Apr 2025 | GitHub: <https://github.com/ajith05/AL-foundation-models>

- Proposed density-based sampling in vision classification which matched baseline accuracy with ~15% fewer labeled samples.

Deep Learning Course Project | Aug 2024 – Dec 2024

- Designed an ensemble movie recommendation model achieving RMSE 0.77 on Netflix Prize dataset.

Machine Learning Course Project | Dec 2020 – Jan 2021

- Built a model to detect forged signatures using 3200+ signatures for training and 720+ signatures for validation.
- Delegated the project tasks to a team of 4 by leveraging members' abilities to meet the tight deadlines.

CERTIFICATIONS

- Associate Cloud Engineer - Google Cloud Platform** (Feb 2026)
- Professional Machine Learning Engineer - Google Cloud Platform** (Jul 2024)
- DeepLearning.AI TensorFlow Developer - DeepLearning.AI / Coursera** (Nov 2020)